code base from scratch. Useful, relevant, and up-to-date functional and technical documentation makes this task a great deal easier.

The automated documentation that comes out of a BDD development process can go a long way toward providing the sort of documentation maintenance teams need in order to be effective. The high-level executable specifications help new developers understand the business goals and flow of the application. Executable specifications at the unit-testing level provide detailed worked examples of how particular features have been implemented.

Maintenance developers working on a BDD project find it easier to know where to start when they need to make a change. Good executable specifications provide a wealth of examples of how to test the application correctly, and maintenance changes will generally involve writing a new executable specification along similar lines or modifying an existing one.

The impact of maintenance changes on existing code is also easier to assess. When a developer makes a change, it may cause existing executable specifications to break, and when this happens, there are usually two possible causes:

- The broken executable specification may no longer reflect the new business requirements. In this case, the executable specification can be updated or (if it's no longer relevant) deleted.
- The code change has broken an existing requirement. This is a bug in the new code that needs to be fixed.

Executable specifications are not a magical solution to the traditional problems of technical documentation. They aren't guaranteed to always be meaningful or relevant—this requires practice and discipline. Other technical, architectural, and functional documentation is often required to complete the picture. But when they're written and organized well, executable specifications provide significant advantages over conventional approaches.

1.4 Benefits of BDD

In the previous sections, we examined what BDD looks like and discussed what it brings to the table. Now let's run through some of the key business benefits that an organization adopting BDD can expect in more detail.

1.4.1 Reduced waste

BDD is all about focusing the development effort on discovering and delivering the features that will provide business value, and avoiding those that don't. When a team builds a feature that's not aligned with the business goals underlying the project, the effort is wasted for the business. Similarly, when a team writes a feature that the business needs, but in a way that's not useful to the business, the team will need to rework the feature to fit the bill, resulting in more waste. BDD helps avoid this sort of wasted effort by helping teams focus on features that are aligned with business goals.